Specialized YOLO11 Detection and Selective Prediction for Cafeteria Waste

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Abstract. Waste-sorting assistance requires more than identifying discarded objects: their appropriate disposal streams depend on material composition, contamination, contents, and local collection rules. We examine a prototype comprising four specialized YOLO11 detectors for seven cafeteria-item classes and a rule-based policy that recommends composting, recycling, or manual review. On an historically exposed development cohort of 248 photographs with 924 AI-assisted object annotations, the frozen detectors achieved 87.78% mAP50 and 67.24% mAP50–95. At a uniform confidence threshold of 0.50, micro precision and recall were 93.42% and 78.35%. Class-specific thresholds selected on 186 validation photographs increased precision to 97.03% on validation and 97.22% on the historical cohort, while reducing recall to 71.43% and 68.18%, respectively. Historical-cohort F1 consequently declined from 85.23% to 80.15%. Only six historical-cohort images were captured at the participating school, and a provenance audit identified neither a confirmed independent test cohort nor verified ground-truth disposal-bin labels. The study therefore establishes a reproducible development-set precision–coverage trade-off, not real-world disposal accuracy or reduced waste contamination.

Keywords: cafeteria waste, YOLO11, object detection, selective prediction, waste classification, evaluation provenance.

1 Introduction

At the end of a school lunch period, a tray may contain leftover food, utensils, a drink carton, and several forms of packaging. Recognizing these items is relatively straightforward to formulate as a computer-vision problem; determining where each belongs is not. A carton may be recyclable only after it has been emptied, while packaging that appears compostable may not be accepted by the local collection program. The U.S. Environmental Protection Agency accordingly emphasizes both the need to consult local recycling guidance and the importance of emptying and cleaning containers [1]. Even a correct visual prediction can therefore lead to an inappropriate disposal recommendation.

Object detection provides a natural foundation for an assistance system because it identifies multiple items and their locations within the same photograph. Yet the confidence assigned to a detection expresses the model's assessment of its visual prediction; it does not certify an object's material, condition, or disposal eligibility. A useful system must distinguish recognition from decision-making and preserve a mechanism for withholding recommendations when the available evidence is insufficient. It must also account for objects that are never detected, since excluding uncertain predictions can make reported precision look stronger while leaving more items unaddressed.

This study evaluates a prototype combining four specialized YOLO11 detectors with a deterministic disposal-suggestion policy. Together, the detectors recognize seven item categories; the policy recommends composting or recycling for selected categories and directs others to manual review. We assess the unchanged checkpoints on available development images, investigate the precision–recall consequences of choosing separate confidence thresholds for each class, and audit the provenance of both images and labels. The central question is not whether the system has already solved cafeteria waste sorting, but how reliably its current visual predictions can support a cautious, auditable recommendation process.

1.1 Related Work

Early waste-recognition systems generally treated the task as whole-image classification. Thung and Yang's TrashNet project classified isolated discarded objects into six material-oriented categories [2]. More recently, Single, Iranmanesh, and Raad introduced RealWaste, using photographs collected in a landfill environment to study recognition under less controlled conditions [3]. The contrast matters: a model trained on cleanly presented objects may face a substantially different visual problem when materials are dirty, crumpled, overlapping, or surrounded by unrelated waste.

Other datasets frame waste recognition as instance localization rather than image-level categorization. Proença and Simões introduced TACO, which provides annotations for litter in its surrounding context and supports instance-segmentation research [4]. Bashkirova and colleagues extended this emphasis on difficult scenes with ZeroWaste, a collection of cluttered recycling-conveyor images containing deformable, overlapping objects [5]. Both illustrate why

mixed waste is poorly represented by a single label per photograph. Neither, however, makes local disposal eligibility equivalent to visual class: a detected plastic object can still require information about its material specification, contents, or collection rules before an appropriate bin can be established.

The detection architecture in this study belongs to the YOLO family introduced by Redmon and colleagues [6] and uses the YOLO11 implementations distributed by Ultralytics [8]. We follow the localization-sensitive evaluation conventions associated with Microsoft COCO, including average precision at multiple intersection-over-union thresholds [7]. These metrics summarize recognition across ranked detections, but they do not specify how a particular confidence cutoff behaves in practice. For an assistance system that acts only on selected predictions, fixed-threshold precision and recall provide complementary information.

The decision to withhold uncertain outputs connects this work to selective classification. Geifman and El-Yaniv examined confidence-based rejection [9] and subsequently introduced SelectiveNet, which integrates a rejection option into neural-network training [10]. Our method is deliberately narrower: it changes confidence cutoffs for existing detectors without retraining them or implementing a learned rejector. Higher precision among the retained predictions must therefore be assessed alongside the objects that become undetected or require review.

Finally, the reliability of any claimed improvement depends on evaluation design. WILDS documents the difficulty of transferring learned models across distributions and collection environments [11], while Kapoor and Narayanan describe how leakage can compromise reproducibility in machine-learning research [12]. These concerns are immediate here: images originate from several datasets, parts of the model-selection history remain incomplete, and the nominal historical test cohort was previously consulted during development. We consequently distinguish between observations supported by the available development records and generalization claims that would require a new, sealed evaluation.

2 Data and Annotation Provenance

The maintained dataset contains 2,372 training views, 186 validation photographs, and 248 photographs in the historical test partition. Of the training views, 2,052 are original image entries and 320 are additional augmented views; twenty school photographs account for 340 views when their augmentations are included. Augmentation increases the variety of presentations seen during training without creating independently captured scenes. Because the specialist checkpoints were developed through different historical training and selection procedures, the current manifest should not be interpreted as the exact common training set for all four models.

The images come from public waste datasets and two school submissions. The historical cohort combines 73 school-lunch dataset images, 66 tetra-pak images, 48 trash-detection images, 31 plate images, 20 utensil images, four macworx images, and six photographs collected at the participating school. Thus, only 2.42% of the historical cohort consists of school-origin photographs. The separate dataset named school_lunch is not part of that school collection, and the cohort cannot be treated as a representative field sample from the cafeteria.

We score detections against the frozen, repaired annotation manifests rather than older label files that no longer reflect the accepted evaluation records. The historical cohort contains 924 annotated objects. A previous validation audit documented discrepancies in 109 older on-disk label files, including ten involving bottle annotations. None of these labels was changed for the present editorial revision; the recorded manifests and hashes remain the authoritative evaluation inputs.

Annotation quality also limits the interpretation of the scores. Historical annotations were produced or reviewed with AI assistance, and the available records do not establish comprehensive independent human verification. The school checklist, for example, contains 123 entries with blank human-review fields. Dense food scenes can also make the boundary between adjacent servings ambiguous. This uncertainty is particularly consequential for hinged containers, represented by just two validation annotations and four in the historical cohort.

Table 1. Annotated objects by class and split. Training counts include augmentation views and are not counts of independent photographed objects; validation and historical-test counts use the frozen repaired annotations.

Class	Training-view boxes	Validation boxes	Test boxes
Food	2,501	343	433
Plate	722	110	123
Utensil	457	101	98
Hinged container	185	2	4
Drink carton	972	94	134
Wrapper	316	85	75

Class	Training-view boxes	Validation boxes	Test boxes
Bottle	293	42	57

3 Model Architecture and Inference

Rather than training one model to recognize every cafeteria item, the prototype combines four previously adapted YOLO11 specialists. Each checkpoint is responsible for a defined subset of the seven output categories, as summarized in Table 2. This division reflects the retained development configuration, not a controlled comparison showing that specialization outperforms a unified detector. The checkpoints, class mappings, inference parameters, and disposal rules were recorded in the project's frozen configuration at commit defcc6156a4425daf1ae530757634cc541f3c267. The protected hashes were checked during the research audit so that the reported comparisons refer to unchanged model artifacts.

Table 2. Frozen specialist checkpoints, parameter counts, class ownership, and per-model inference resolution.

Specialist	YOLO11 variant	Parameters	Input size	Owned classes
food-container-carton	yolo11n.yaml	2,590,425	640	food, hinged_container, drink_carton
plate-wrapper	yolo11n.yaml	2,605,126	640	plate, wrapper
utensil	yolo11x.yaml	56,877,241	800	utensil
bottle	yolo11s.yaml	9,428,179	640	bottle

Inference was performed with Ultralytics 8.4.175, PyTorch 2.14.1, and Python 3.12. Each specialist processed one image at a time, without test-time augmentation, using its configured input resolution. Non-maximum suppression used an IoU threshold of 0.70 and a maximum of 300 detections per specialist. The saved protocol also retains half=False, an option deprecated in the installed library, to document the original invocation. Predicted boxes were represented in normalized XYXY coordinates. After each specialist's native suppression, outputs were concatenated by their assigned classes; no further cross-specialist deduplication was performed.

Evaluation took place on an Apple M1 MacBook Air with 8 GiB of memory and MPS support. These details identify the recorded computing environment rather than establish efficiency: latency, energy use, and throughput were not measured. Moreover, the four checkpoints have different adaptation and selection histories, so their combined performance should not be described as the outcome of a single controlled training experiment.

Detection outputs were retained down to confidence 0.001 for average-precision calculations. For the default display policy, predictions at or above 0.25 were shown; predictions below 0.50 were referred for review. At confidence 0.50 or above, the proposed disposal stream followed the class-dependent rules in the table below.

Detected category	Default suggestion at confidence 0.50
Food	Compost
Bottle or drink carton	Recycling
Plate, utensil, wrapper, or hinged container	Manual review

These rules encode a provisional decision policy, not independently verified disposal eligibility. In particular, the recycling suggestion does not establish that a container is empty or accepted by the school's collection program. The system does not issue an automatic landfill recommendation.

4 Evaluation Methods

We evaluate localization and classification jointly. Intersection-over-union (IoU) measures the overlap between a predicted bounding box and its reference annotation, divided by the area of their union. The primary ranking-based measures are mAP50, averaged across the seven classes at IoU 0.50, and mAP50–95, which additionally averages over thresholds from 0.50 to 0.95 in steps of 0.05 [7]. Both are calculated from confidence-ranked detections using the retained Ultralytics evaluation procedure; neither represents precision at one operational cutoff.

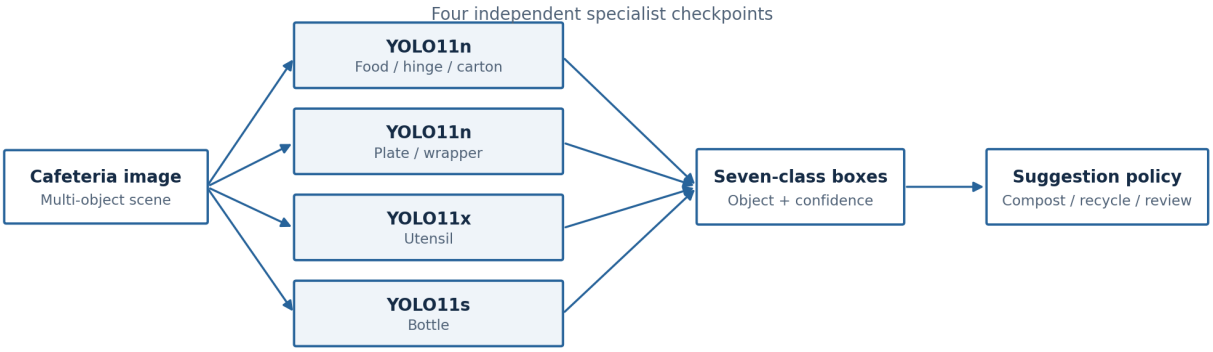


Figure 1: Processing pipeline: four YOLO11 specialists produce seven item classes, which are mapped to provisional compost, recycling, or manual-review suggestions.

For fixed-threshold evaluation, detections are considered in descending confidence order and matched one-to-one to unused annotations of the same class when IoU is at least 0.50. Matched detections count as true positives (TP), unmatched detections as false positives (FP), and unmatched annotations as false negatives (FN). Aggregating these counts over all classes gives micro-averaged precision, recall, and F1:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}.$$

We distinguish these detection measures from two object-level proxies for automatic disposal suggestions. The first counts class-consistent matches separately for food, bottle, and drink carton. The second assigns displayed detections to reference boxes by geometry before checking their classes, thereby exposing competition between overlapping predictions from different specialists. A prediction that matches a food box but conflicts with a wrapper annotation illustrates why these approaches can yield different totals. Neither proxy can establish whether the object actually belongs in the suggested bin.

The evaluation also distinguishes development evidence from independent generalization. The 248-image historical cohort was examined during earlier model development and selection, so it cannot be treated as a sealed holdout. Class-specific thresholds were determined from the separate 186-image validation cohort and then applied unchanged to those historical images. Since thresholding only filters an existing ranked set of predictions, it changes fixed-cutoff precision and recall but does not change the frozen detectors' mAP. Previous use of evaluation data can otherwise inflate apparent performance [12].

5 Confidence Threshold Selection

The initial experiment tested whether one confidence cutoff could improve precision while preserving useful detection coverage. On the 186-image validation cohort, we evaluated global thresholds from 0.50 to 0.99 at 0.01 intervals, together with 0.995. No single threshold simultaneously achieved the three prespecified criteria of at least 97% micro precision, at least 60% recall, and at least 100 accepted predictions. Among the global cutoffs satisfying the latter two requirements, the highest precision was 96.04% at a threshold of 0.79, with 68.73% recall.

We then allowed each class to have its own confidence threshold. Candidate cutoffs for each class were summarized by their validation-set true- and false-positive counts. A dynamic program combined these classwise choices, preserving the largest attainable number of true positives for each feasible false-positive total. The final threshold vector maximized correctly accepted detections subject to the same precision, recall, and accepted-prediction constraints:

$$\max_{\tau} TP(\tau) \quad \text{subject to} \quad P(\tau) \geq 0.97, \quad R(\tau) \geq 0.60, \quad N_{\text{pred}} \geq 100.$$

Table 3. Class-specific confidence thresholds selected on development validation.

Food	Plate	Utensil	Hinged container	Drink carton	Wrapper	Bottle
0.79	0.83	0.50	0.50	0.84	0.83	0.91

The selected vector was subsequently applied without modification to the historical cohort. It was not used to replace the existing disposal-policy settings, and no additional cutoff selection was performed on those historical images. Because those images had already informed development, the resulting comparison remains descriptive rather than an independent estimate of field performance.

6 Results

Table 4. Per-class fixed-threshold object-recognition metrics on the exposed 248-photo cohort (confidence 0.50, IoU 0.50); AP uses the full frozen score range. These are descriptive development results, not independent bin classification.

Class	TP / FP / FN	Precision	Recall	F1	AP50	AP50–95
food	327 / 10 / 106	97.03%	75.52%	84.94%	83.91%	55.09%
plate	103 / 15 / 20	87.29%	83.74%	85.48%	87.04%	71.60%
utensil	70 / 7 / 28	90.91%	71.43%	80.00%	86.06%	61.26%
hinged_container	4 / 1 / 0	80.00%	100.00%	88.89%	99.50%	91.17%
drink_carton	129 / 1 / 5	99.23%	96.27%	97.73%	97.55%	74.28%
wrapper	58 / 7 / 17	89.23%	77.33%	82.86%	85.13%	57.64%
bottle	33 / 10 / 24	76.74%	57.89%	66.00%	75.27%	59.62%

The historical cohort achieved 87.78% mAP50 and 67.24% mAP50–95; the corresponding validation scores were 86.36% and 65.25%. These macro-averages give each class equal weight, regardless of its annotation count. The hinged-container AP50 of 99.50%, for example, is based on only four historical-cohort objects and should not be interpreted as robust performance for that category. At the uniform 0.50 confidence cutoff, pooled detection precision, recall, and F1 on the historical cohort were 93.42%, 78.35%, and 85.23%, respectively. Detailed validation results are available in the frozen evaluation records.

Table 5. Automatic-suggestion object proxies, explicitly separate from bin accuracy. Recall uses the 624 food/bottle/carton object labels.

Matching convention	Correct object matches	Precision	Recall	F1
Class-specific	489 of 510 automatic suggestions	95.88%	78.37%	86.24%
Joint geometry-first	483 of 510 automatic suggestions	94.71%	77.40%	85.19%

Class-specific matching counted 489 correct object detections among 510 automatic suggestions; geometry-first matching retained 483. Five food detections and one bottle detection were instead assigned to overlapping wrapper labels under the latter convention. These are disagreements in object assignment, not verified wrong-bin outcomes. The frozen 248-image predictions were reproduced in a separate inference replay, and the present analysis uses those saved outputs without retraining the models.

Table 6. Default sorting-policy outputs at the 0.25 display-confidence floor on the exposed historical development cohort; results are not independently verified bin assignments.

Quantity	Count / denominator	Value
Compost suggestions	337 / 881 emitted detections	38.25%
Recycling suggestions	173 / 881	19.64%
Automatic suggestion rate	510 / 881	57.89%
Manual-review rate among emitted detections	371 / 881	42.11%
Geometry-matched automatic coverage of labeled objects	489 / 924	52.92%

Quantity	Count / denominator	Value
Exact-class correct automatic matches / labeled objects	483 / 924	52.27%
Geometry-matched review / labeled objects	286 / 924	30.95%
Missed objects	149 / 924	16.13%
Unmatched automatic predictions	21 / 510 automatic suggestions	4.12%
Unmatched review predictions	85 / 371 review predictions	22.91%

At the display threshold of 0.25, the default policy emitted 881 detections: 337 compost suggestions, 173 recycling suggestions, and 371 manual-review outcomes. Automatic suggestions therefore accounted for 57.89% of emitted detections. This denominator is narrower than the set of actual objects in the photographs, since undetected items generate no suggestion or review record. Indeed, exact-class automatic matches account for 483 of the 924 annotated objects (52.27%), a coverage-sensitive detection proxy rather than correct-bin accuracy. We do not attach conventional object-independent confidence intervals to these results: predictions are clustered within photographs, annotations contain unresolved uncertainty, and the historical cohort has prior development exposure.

6.1 Precision–Recall Trade-off After Threshold Selection

Table 7. Overall seven-class micro object recognition at box IoU ≥ 0.50; same source images and labels within each cohort.

Cohort / policy	TP	FP	FN	Precision	Recall	F1
Validation (186), uniform 0.50	638	48	139	**93.00%**	82.11%	87.22%
Validation (186), classwise selected	555	17	222	**97.03%**	71.43%	82.28%
Historical (248), uniform 0.50	724	51	200	**93.42%**	78.35%	85.23%
Historical (248), classwise fixed	630	18	294	**97.22%**	68.18%	80.15%

The classwise policy reduced historical-cohort false positives from 51 to 18, but also removed 94 true positives, from 724 to 630. The number of accepted predictions declined from 775 to 648. Consequently, precision increased by 3.80 percentage points while recall decreased by 10.17 points; F1 fell from 85.23% to 80.15%. Validation F1 likewise fell from 87.22% to 82.28%. The result supports a specific trade-off—fewer false-positive detections among accepted outputs—rather than an overall improvement in recognition quality. Neither the model weights nor the ranking-based mAP changed.

7 Independence and Annotation Audit

A reliable evaluation requires photographs that have not influenced training, threshold selection, or earlier model decisions. The project's local inventory was therefore screened for duplicate files, overlapping capture groups, and recorded prior inference. The audit covered 102,806 readable image paths, corresponding to 67,422 distinct filesystem inodes and 14,379 byte-distinct images. Within the 9,468 candidate image identities, 3,842 were flagged for possible exposure, group overlap, or duplication; 5,482 had unresolved historical provenance; and 144 original HEIC files

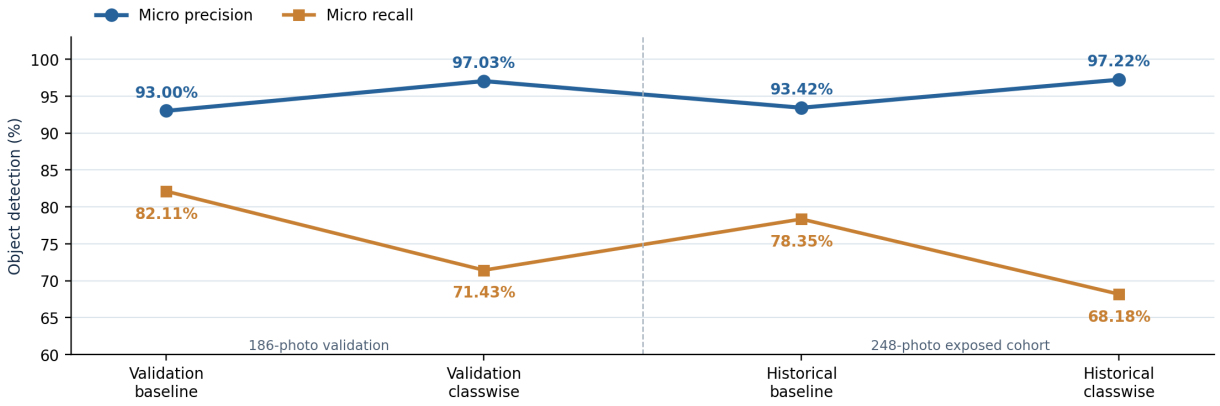


Figure 2: Precision–recall trade-off for default and validation-selected classwise confidence thresholds, measured on exposed development cohorts.

could not be visually decoded in the audit environment. No image was *confirmed* independent under the available evidence (Table 8). This does not prove that every candidate was used for model fitting.

The audit compared SHA-256 file identities and decoded-pixel hashes and screened potential near-duplicates using rotation/reflection-invariant comparisons of 64-bit difference hashes against 4,751 images from recorded training directories or prepared manifests. A Hamming-distance threshold of five was used for screening. These tests are deliberately conservative: membership in a training directory is not proof of optimizer exposure, while the absence of a perceptual match cannot rule out related crops, viewpoints, or capture sessions.

Table 8. Provenance status of the 9,468 candidate image identities. The categories reflect what could be established from available records, not a definitive partition of images actually used for training.

Candidate status	Image identities
Exposure, group-overlap, or duplicate flag	3,842
Unresolved historical provenance	5,482
Original HEIC visual decoding unavailable	144
Confirmed independent	0

The school-specific history also prevents a prospective test claim. All 123 photographs in the earlier school submission and all 21 in the later submission shared recorded training or development capture groups. Four images from the October 29 submission entered the training collection, so that capture day was excluded from consideration as an independent test source. Earlier labels such as "sealed test" are superseded by subsequent prediction receipts and model-use records. Several audit leads overlap and must not be added together as unique training exposures.

Annotation provenance is a separate limitation. The metadata audit searched 2,398 JSON files and additional CSV, XML, and YAML files but found no confirmed set of independently human-adjudicated, per-object disposal-bin labels. A historical object-review flag is not a substitute for an actual material and disposal-eligibility annotation. The audit also noted 42 unreadable image aliases and five malformed or inaccessible JSON records, which were retained rather than silently repaired. Full provenance diagnostics and hashes are preserved in `results/research-readiness.json` and the external audit artifacts.

Table 9. Real-world sorting metrics that cannot be computed from the current development annotations.

Metric	Status	Evidence required
Precision, recall, and F1 for actual disposal bins	Not measured	Independent images and adjudicated bin labels
Correct-bin accuracy	Not measured	Verified local disposal eligibility
Automatic coverage of independently labeled objects	Not measured	Complete, independent object annotations
Manual-review burden on new cafeteria photographs	Not measured	A sealed school cohort

The lack of independently verified bin outcomes is not equivalent to a zero score. It means that the necessary ground truth does not yet exist in the audited records.

8 Error Analysis and Exploratory Stress Test

The diagnostic decomposition uses frozen object labels and predictions, without editing annotations or changing thresholds. A false-negative label is classified as below threshold if a same-class box has IoU>=0.50 but score<0.50; a localization candidate has same-class IoU>=0.10 but <0.50; otherwise it has no same-class overlap at that cutoff and saved confidence floor. These are competing hypotheses, not human-confirmed causes. A box can overlap other objects or depend on uncertain serving definitions. Background-or-missing-annotation likewise does not distinguish model hallucination from an incomplete label set.

Table 10. Frozen historical-cohort object-detection error diagnostics for food and bottle at the default 0.50 confidence cutoff. Error types are algorithmic hypotheses, not human-confirmed causes.

Diagnostic category	Food	Bottle
Correct class matches at automatic threshold	327	33
FN: below automatic threshold	53	19
FN: localization candidate	48	1
FN: no same-class overlap	5	4
FP: localization candidate	2	0
FP: duplicate/competing detection	1	0
FP: background or missing annotation	7	9
FP: other-class overlap/annotation ambiguity	0	1

Errors were concentrated in a small number of difficult scenes. One buffet photograph accounted for 52 of the 106 missed food objects (49.06%), while a single outdoor-trash photograph accounted for 10 of the 24 missed bottles (41.67%). Multiple errors in one scene should not be interpreted as independent failures across many photographs. Both scenes remained in the reported results.

AI visual inspection covered 12 deterministically selected error cases from 10 unique photographs; it is not a random audit or a human-label correction. Examples suggest a reusable-looking sports bottle outside the disposable-bottle scope, a possible missing bottle annotation, small cylindrical drink-package class ambiguity, dense buffet scale/serving-granularity problems, and sealed food packaging detected as food. Actual material, contents and bin acceptance cannot be settled from these previews. No labels or sorting rules were changed.

A separate, previously completed bottle-specialist experiment tested one additional epoch of classification-head training. Validation AP50 declined from 82.47% to 77.95%, and AP50–95 fell from 62.00% to 59.18%. Because the candidate failed the predefined validation criterion, it was rejected without being activated or evaluated on the historical test cohort. The frozen system reported here was not retrained during manuscript finalization.

8.1 Exploratory Public-Image Stress Test

After the research freeze, the unchanged four detectors were run on 44 source-screened public photographs (nine bottle-source, six carton-source, 15 utensil-source, 12 hinged-container-source and two wrapper-source images). There were 149 displayed predictions at the 0.25 confidence floor: 12 compost suggestions, 49 recycling suggestions and 88 manual-review outcomes. Three photos had no displayed detections. These counts describe emitted predictions, not verified outcomes: there are no human-verified object boxes, documented capture-day independence, or school bin-eligibility labels. Accordingly, no **human-verified detection mAP, correct-bin accuracy, or sorting precision** is available for the complete 44-image cohort. The separate 41-image approximate-label diagnostic below is not a substitute for that missing evaluation.

A preliminary image-level *source-hint* cross-check found at least one matching model class on 9/9 bottle-source, 5/6 carton-source, 12/15 utensil-source, 4/12 hinged-container-source and 0/2 wrapper-source photos. These fractions are **not recall**: source names are not exhaustive human object labels, and some images contain multiple target items. Visual inspection suggests repeated plate/hinged-container confusion on open foam food boxes and missing wrapper detections. A foil meal tray was detected as `drink_carton` at confidence 0.617, triggering an automatic recycling suggestion under the frozen user-selected policy. This is a plausible wrong-object automatic action; actual local bin eligibility cannot be concluded from the photo alone. A carton photo with three cartons yielded four

carton boxes and another carton image yielded none. Crowded bottle images also exhibited overlapping boxes and cross-specialist labels.

A separate provisional comparison scored 41 of these 44 public-source photographs using approximate AI-generated object labels. The unchanged V1 detectors obtained 46.25% mAP50 and 21.16% mAP50–95; a developmental hinged-container V3 candidate obtained 44.78% mAP50 and 21.47% mAP50–95. These numbers are not comparable to a sealed test benchmark: the annotations were not independently human-verified, three crowded bottle photographs were excluded from the scoring subset, and the entire source set had already been inspected during development. The V3 candidate was not adopted as the frozen system. This comparison is included only to document an exploratory stress test, not to claim a measured out-of-distribution performance change.

These observations describe possible failure modes, not independently measured detection or disposal accuracy, and were not used to authorize a new training run. Individual photo references and limitations are documented in the visual audit. These public images may be out-of-distribution relative to actual cafeteria disposal scenes, and original source rights require review before reproducing them in a publication.

9 Limitations and Future Evaluation

The main limitation of this study is that the reported benchmark images are development data, not a sealed holdout. As demonstrated in research on distribution shift and leakage, performance on reused or mismatched source data may not transfer to the intended deployment setting [11, 12]. Earlier inference, model selection, and related capture groups prevent an independent generalization claim. The object annotations were also AI-assisted, with limited evidence of human verification; in particular, the hinged-container class has only two validation objects and four historical-test objects. Food-serving boundaries and visually similar packaging introduce further annotation uncertainty. The existing audit reduces uncertainty about what data and model versions were used, but incomplete historical records mean that the absence of a detected duplicate cannot establish independence.

A second limitation concerns the distinction between an object class and a disposal decision. The current rule layer does not inspect whether a bottle or carton is empty, whether food is contaminated by packaging, or whether a specific material is accepted by the school's collection program. It cannot automatically assign landfill disposal, and requesting manual review is an abstention rather than a correct sorting outcome. The results therefore cannot quantify correct-bin accuracy or reductions in waste contamination. Latency, throughput, energy use, and user interaction during a real cafeteria deployment have also not been measured. Source-license and attribution records must be checked before any external image redistribution.

The next evaluation should use newly collected cafeteria photographs from capture sessions excluded from all training, threshold selection, and prior inspection. Inclusion criteria and capture-group membership should be fixed before model inference. Annotators who have not seen the model's predictions should mark each relevant object's bounding box and actual disposal eligibility using documented local collection rules and, when necessary, material and contents information unavailable from the photograph. Uncertain cases should be adjudicated or recorded as unscorable under a predefined protocol. On that sealed cohort, the unchanged system can then be evaluated for object detection, correct-bin precision and recall, prediction coverage, and manual-review burden, with uncertainty estimated across independent capture groups rather than treating multiple objects in one photograph as unrelated observations. For each disposal stream, an automatic wrong-bin recommendation should count against the predicted stream; a reviewed or missed item should not be credited as a successful automatic disposal decision. Reporting only accuracy among accepted decisions would conceal the effect of abstention, so the study should also report the fraction of *all* eligible labeled items assigned correctly without review.

10 Conclusion

The four-specialist YOLO11 system achieved 87.78% mAP50 and 67.24% mAP50–95 on 248 historically exposed development photographs. Selecting class-specific thresholds on validation increased micro object-detection precision to 97.03% on that cohort and 97.22% on the exposed historical cohort, but reduced recall and F1. This result demonstrates a controllable precision–coverage trade-off, not a general improvement in recognition or a verified ability to direct cafeteria waste to the correct disposal stream. The next decisive experiment is evaluation of the frozen detector and disposal policy on new, independently annotated school photographs with ground-truth bin eligibility.

11 Reproducibility and Data Availability

The retained implementation, experiment configuration, and summary results are maintained in the GitHub repository. The paper's frozen configuration, `configs/research-freeze.json`, records the four checkpoint hashes, class mappings, preprocessing settings, and active disposal rules. Existing benchmark outputs are in `results/sorting-verification.json`, with provenance and annotation limitations in `results/research-readiness.json`. The corresponding procedures are documented in `docs/sorting-evaluation.md`.

The underlying images, checkpoint binaries, and detailed local audit artifacts are stored separately from Git in `~/Desktop/yolo-cafeteria-research-assets/`. In particular, the combined dataset manifest and the repaired validation and historical-test annotations are external assets. Source URLs and historical licensing declarations are recorded in `data/datasets.csv` and `data/manifests/` within that asset collection, but licenses and redistribution rights have not been independently verified. The GitHub source alone is therefore insufficient to reproduce native inference, and no public release of the collected school photographs is implied.

The original audit implementation is retained in `scripts/research_audit.py`. With the external asset environment present, its file inventory can be reproduced by executing:

```
../yolo-cafeteria-research-assets/venv/bin/python -B scripts/research_audit.py --output /tmp/cafeteria-asse
../yolo-cafeteria-research-assets/venv/bin/python -B -m unittest discover -s tests -v
```

These commands audit retained files and run software tests; they do not retrain YOLO or establish independent sorting accuracy. A future bin-label evaluation should use `scripts/evaluate_sorting.py` with a new, independently verified disposal-eligibility annotation file and the frozen prediction policy.

12 References

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